

LABOR DEMAND ON A TIGHT LEASH

MARIO BOSSLER AND MARTIN POPP*

Using detailed information on vacancies and job seekers, the authors study the effect of labor market tightness on labor demand for the near-universe of German firms. To this end, novel Bartik instruments are constructed that combine firms' predetermined employment shares with nationwide shifts at the occupational level. The results show that tightness significantly reduces firms' labor demand, implying that the observed doubling in tightness between 2012 and 2019 reduced employment by 5%. At the aggregate level, the negative tightness effect creates search externalities, which reduce the own-wage elasticity of labor demand from -0.7 to -0.5 through reallocation of workers between firms. To guide the analysis, the authors embed elements of the canonical search-and-matching model into a labor demand equation, while allowing vacancy posting costs to increase in tight markets. Through the lens of this model, the pre-match component of hiring costs amounts to 16–24% of annual wage payments.

Over the past decade, the German economy experienced a remarkable upswing. Between 2012 and 2019, Germany's real gross domestic product grew on average by 1.6% each year. At the same time, the German labor market witnessed the biggest expansion since the 1950s: The number of jobs rose by 3.9 million, reaching a record high of 45.4 million in 2019. As a flip side of this so-called German Labor Market Miracle (Burda and Seele 2020), labor market tightness—the ratio of vacancies to job seekers—doubled. As a consequence of the increased scarcity of labor, German businesses lamented the lack of workers. However, evidence on the extent to which the tightening has prevented firms from retaining or expanding

*MARIO BOSSLER (<https://orcid.org/0009-0006-6132-7463> ) is a Professor at the Nuremberg Tech University of Applied Sciences. MARTIN POPP (<https://orcid.org/0000-0002-9097-9798> ) is a Postdoctoral Researcher at the Institute for Employment Research (IAB).

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their workforce is missing. Quantifying the employment effects of tightening labor markets would also prove insightful since many industrialized economies have been facing labor shortages in recent years (e.g., Abraham, Haltiwanger, and Rendell 2020 for the United States).

In tight labor markets, firms compete to fill a relatively large number of vacancies from a relatively small number of job seekers. High levels of tightness give rise to hiring frictions, making it expensive for firms to recruit workers. Although this channel is key, empirical models of labor demand narrow their analysis to the wage rate and do not consider the role of labor market tightness. The paucity of empirical evidence on the effect of labor market tightness on employment is twofold: On the one hand, detailed information on both vacancies and job seekers per labor market is rarely available. On the other hand, failure to isolate exogenous variation in labor market tightness would lead to spurious estimates.

In this article, we estimate the causal effects of not only wages but also labor market tightness on labor demand in German firms. To provide microfoundations for our empirical approach, we integrate components of the canonical search-and-matching model into a labor demand equation and, unlike standard practice, allow vacancy posting costs to increase in labor market tightness. The vacancy posting costs, which refer to the pre-match component of hiring costs, drive a wedge between unit labor cost and the wage rate. This framework allows us to derive an estimable labor demand equation which, in contrast to conventional specifications, captures a negative effect of labor market tightness as hiring becomes more costly. We estimate this equation on the universe of social security records from the German labor market between 2012 and 2019. For the purpose of our analysis, we enrich these data with official statistics and survey information on vacancies and job seekers to determine firm-specific exposure to labor market tightness in more than 1,200 occupations.

A naive ordinary least squares (OLS) regression of firms' labor demand on wages and labor market tightness would provide upward-biased elasticities. To rule out bias from uncontrolled shifts in labor demand, we instrument wages and labor market tightness with shift-share instruments, building on the popular instrumental variable (IV) strategy from Bartik (1993). Typically, Bartik instruments combine national industry shifts with past shares of industries in regions to isolate exogenous variation in variables at the regional level. However, we transfer this shift-share design to the firm level by taking advantage of the fact that a firm employs many occupations, just as a region has many industries. Thus, our novel Bartik-style instruments combine national occupation shifts with past shares of occupations in firms' employment to extract plausibly exogenous variation at the firm level.

Our microfounded model, in combination with our empirically estimated elasticities, allows us to gain two additional key insights into the interplay between labor demand, wages, and labor market tightness. First, the model highlights that the negative effect of tightness on labor demand gives rise to search externalities: A reduction in labor demand by one firm lowers labor market tightness and, thus, facilitates the recruitment of workers in all other firms, leading to reallocation effects of workers between firms. Guided by our model, we quantify the relevance of these reallocation effects. Second, owing to the manifold search-and-matching ingredients in our model, our estimated elasticities indirectly inform us about the importance of vacancy posting costs. Specifically, after calibrating only a few parameters, we insert the estimated wage and tightness elasticities of labor demand in our model to gauge the relative magnitude of pre-match hiring costs as a fraction of annual wage payments.

Embedding in the Literature

Our article contributes to several strands of the literature. First, we join the proliferation of studies that attempt to estimate the own-wage elasticity of labor demand (Hamermesh 1986; Nickell 1986). A major concern in these studies is the endogeneity of wages, namely that unaccounted shifts of the labor demand curve will yield upward-biased elasticities. The use of quasi-experimental variation in wages represents a promising method to address problems of endogeneity (Addison, Portugal, and Varejão 2014). Unfortunately, quasi-experimental studies often lack external validity by focusing on rather narrow policy designs (e.g., low-wage workers when studying variation in minimum wages). Our novel Bartik-like instruments are designed to isolate plausibly exogenous variation at the firm level without requiring us to restrict the analysis to specific groups of workers or submarkets. Since our shift-share design rigorously addresses upward bias, our estimated own-wage elasticity is at the lower end of the values reported in the international and German literature on labor demand (Lichter, Peichl, and Siegloch 2015; Popp 2023).

Second, we add to the small but growing literature studying the consequences of the scarcity of labor inputs on firms' labor demand. Several studies exploit commuting policies to examine the impact of positive labor supply shocks on domestic employment in border regions (Dustmann, Schönberg, and Stuhler 2017; Beerli, Ruffner, Siegenthaler, and Peri 2021). Other work builds on business cycles to analyze whether the availability of workers affects firms' skill requirements (Modestino, Shoag, and Ballance 2016, 2020; Hershbein and Kahn 2018). While studying labor supply shifts is informative per se, it constitutes only a partial measure of the scarcity of the labor input by neglecting the labor demand side. To infer scarcity of the labor input, researchers should ideally employ the vacancy-to-

job-seeker ratio (i.e., labor market tightness) or, alternatively, use an equilibrium outcome as a proxy thereof.¹

Relying on both supply and demand forces, Beaudry, Green, and Sand (2018) performed the most comprehensive analysis to estimate the causal impact of labor market tightness on employment. Building on an extended search-and-matching framework, the authors leveraged census data on the US economy between 1970 and 2015 and estimated elasticities at the city level using conventional Bartik instruments. Building on 10-year differences, the study found that a 10% increase in the employment rate (as a proxy for labor market tightness) reduces aggregate employment by approximately 20%. From an empirical point of view, our article extends the aforementioned study by analyzing the effects of the tightness variable itself (differentiating between 1,286 occupations at the 5-digit level) on employment at the firm level (while focusing on 2-year differences). Despite conceptual differences and an alternative setting, our findings resonate with Beaudry et al. (2018) in the sense that reduced availability of the labor input is detrimental to employment.

Third, our article also speaks to the literature on hiring costs. Whereas business surveys provide direct evidence on hiring costs (e.g., Oi 1962), dynamic labor demand models indirectly infer the size and shape of these costs using regression techniques (e.g., Nickell 1986). The most detailed evidence from business surveys implies that pre-match hiring costs sum up to nearly 10% of annual wage payments (Blatter, Muehlemann, and Schenker 2012; Muehlemann and Pfeifer 2016; Muehlemann and Strupler Leiser 2018). However, there are two reasons why even highly specialized surveys may underestimate the true magnitude of pre-match hiring costs. On the one hand, it is difficult to inquire about quasi-fixed costs of hiring (e.g., the expenditure for renting offices for the human resource department or for participating in job fairs), which are frequently inferred to be large (Hamermesh 1989; Abowd and Kramarz 2003; Nilsen, Salvanes, and Schiantarelli 2007). On the other hand, survey information on successful hiring processes is positively selected because it tends to disregard the more difficult and, hence, more expensive recruitment processes, which were either unsuccessful or not undertaken at all. Our indirect approach is able to capture the pre-match component of hiring costs in their entirety, covering not only successful but also unsuccessful recruitment processes. Based on observed labor demand responses on the universe of German firms, our estimate for pre-match hiring costs turns out to be approximately twice as

¹A number of studies use firm-level proxies for tightness (such as vacancy rates or self-reported shortages) to study the impact on firms' outcomes, such as productivity (Haskel and Martin 1993), labor contracts (Fang 2009; Healy, Mavromaras, and Sloane 2015), or investments and capacity utilization (D'Acunto, Weber, and Yang 2020). In a pioneering study on 595 firms, Stevens (2007) showed that employment adjusts more slowly when industry-wide shortages of skilled labor are reported. Compared to labor market tightness, these firm-level proxies are usually inferior measures because they are plausibly related to firms' conduct and, therefore, more susceptible to endogeneity.

large as the evidence from direct business surveys. Moreover, our estimate falls in the lower range of indirect estimates for hiring costs from dynamic labor demand models, which show a wide array of results and range from values near zero (Hall 2004; Asphjell, Letterie, Nilsen, and Pfann 2014) to more than one year of wage payments (Rota 2004; Bloom 2009; Yaman 2019).

Fourth, this article also contributes to the literature on reallocation effects of input factors. Several articles highlight the role of reallocation of workers between occupations (Kambourov and Manovskii 2009; Carrillo-Tudela and Visschers 2023), industries (Golan, Lane, and McEntarfer 2007), and firms (Davis and Haltiwanger 1992; Foster, Grim, and Haltiwanger 2016). In his seminal contribution, Hamermesh (1993) argued that changes in firms' employment overstate aggregate changes in employment to the extent that workers are reallocated between firms within the same aggregate. In recent work, Dustmann et al. (2022) analyzed the 2015 minimum wage introduction in Germany and attributed their finding of close-to-zero employment effects to the fact that many laid-off workers were reallocated to other firms. Using the tightness channel, we find that search externalities noticeably dampen aggregate changes in labor demand by virtue of reallocation effects.

Theoretical Model

We begin with examining the theoretical relationship between wages, labor market tightness, and firms' labor demand to facilitate the later interpretation of our empirical results. To this end, we feed ingredients of the canonical search-and-matching model into a traditional labor demand equation. This approach closely follows Beaudry et al. (2018) whose model we are refining in several ways. For more details, we refer the reader to Online Appendix A, which provides a full derivation of the model.

Baseline Model

Labor Demand Equation

Consider a representative firm in commuting zone z that seeks to maximize profits in a static environment. The firm combines labor L and capital K to produce a homogeneous good. To ease presentation, we postulate a Cobb-Douglas production function, $Y = A \cdot L^{\alpha_1} \cdot K^{\alpha_2}$, with total factor productivity A and constant (or decreasing) returns to scale, $\alpha_1 + \alpha_2 \leq 1$. While we assume that the homogeneous good of production and capital are traded on a national and perfectly competitive market at price P and rate R , we explicitly model that the market for labor is frictional. To facilitate the derivation, we follow Beaudry et al. (2018) and use a metaphor in which firms buy labor services at unit labor cost w through competitive recruitment agencies that face search (or recruitment) frictions when hiring workers.

Profit maximization of the firm implies that unconditional labor demand (in logs) reads as follows:

$$(1) \quad \ln L^* = \frac{1}{1 - \alpha_1 - \alpha_2} \cdot \ln \left(\alpha_1^{1-\alpha_2} \cdot \alpha_2^{\alpha_2} \right) - \frac{1 - \alpha_2}{1 - \alpha_1 - \alpha_2} \cdot \ln w \\ + \frac{1}{1 - \alpha_1 - \alpha_2} \cdot \ln (P \cdot A \cdot R^{-\alpha_2})$$

Thus, optimal labor demand of the firm negatively depends on unit labor cost, while it positively depends on the product price, productivity, and the rental rate of capital.

Job-Creation Curve

The derivation of Equation (1) remains agnostic about the components underlying the unit labor cost w for the representative firm. In a frictionless and thus perfectly competitive market, unit labor costs comprise only the market wage, which is formed by supply and demand. In fact, however, frictions usually hinder the smooth functioning of labor markets and render recruitment for firms costly. In the next step, we explicitly model that the representative firm is impacted by search (or recruitment) frictions in the labor market through buying labor services from recruitment agencies that hire workers in frictional markets.

We postulate that there is a continuum of recruitment agencies with an internal discount rate r . Each of the recruitment agencies can post a vacancy at any point in time to recruit a worker. When posting, the recruitment agency incurs vacancy posting cost χ per period, which can be best thought of as the pre-match component of hiring costs. The matching between vacancies of the recruitment agencies and job-seeking workers materializes on the basis of a constant-returns-to-scale matching function $M_z = U_z^\nu \cdot V_z^{1-\nu}$ where M_z is the regional number of matches (or successful hires), V_z is the regional number of vacancies, and U_z is the regional number of unemployed. The elasticity of matches with respect to job seekers is given by: $0 < \nu < 1$. Under this matching technology, the recruitment agencies' job-filling rate q is

$$(2) \quad q(\theta_z) \equiv \frac{M_z}{V_z} = U_z^\nu \cdot V_z^{-\nu} = \left(\frac{V_z}{U_z} \right)^{-\nu} = \theta_z^{-\nu}$$

where θ_z is regional labor market tightness: $\theta_z = \frac{V_z}{U_z}$. When the vacancy is filled, the recruitment agency offers labor services to the representative firm at a rate w . As long as the hired worker is employed, the worker receives a wage W per period from the recruitment agency. We abstract from involuntary layoffs and model an exogenous separation rate δ at which matches break up.

Note that the overall hiring costs, which we will denote by C , is the product of the monetary cost for vacancy posting per period, χ , and the duration

of the recruitment process, given by the inverse of the agency's job-filling rate $q(\theta_z)^{-1}$, which equals θ_z^ν . Hence, overall hiring cost is: $C = \chi \cdot \theta_z^\nu$.

Profit maximization of the recruitment agency in combination with a free-entry condition yields the following job-creation curve:

$$(3) \quad w = W + (r + \delta) \frac{\chi}{q(\theta_z)} = W + (r + \delta) \cdot \chi \cdot \theta_z^\nu$$

The job-creation curve implies that unit labor costs for the representative firm (which corresponds to the price of labor services of the recruitment agency) comprise the wage costs W and the so-called amortized hiring costs, $(r + \delta) \cdot \chi \cdot \theta_z^\nu$. Alongside wage costs, profit maximization of the recruitment agency requires that the price for labor services per period must cover only a fraction $(r + \delta)$ of overall hiring costs (Hamermesh 1993).

Overall, the presence of vacancy posting costs drives a wedge between the unit labor cost and the wage rate. In the absence of vacancy posting costs, $\chi = 0$, hiring would be costless and unit labor cost would be equivalent to the wage rate: $w = W$. The importance of vacancy posting costs depends on the firms' discount rate, the separation rate, and the expected search duration of the firm. In particular, the search duration is an increasing function of labor market tightness: When labor markets tighten, there are more competing vacancies and fewer available unemployed persons and, thus, it takes longer for the firm to successfully fill a vacancy.

Vacancy Posting Costs

Following Yashiv (2006), we decompose unit hiring costs into a pre-match and a post-match component. Pre-match hiring costs comprise all search costs of filling a vacancy with a suitable candidate. These costs include expenditures for job advertisement, posting vacancies, screening candidates, interviews, headhunters, signing bonuses, negotiations, or the maintenance of a human resource department. Post-match hiring costs involve all costs after the contract is signed, namely the costs of onboarding new workers. The canonical search-and-matching model solely focuses on the pre-match component of hiring costs by modeling the costs of vacancy posting.² Throughout the article, we use the terms "vacancy posting costs" and "pre-match hiring costs" synonymously.

In the search-and-matching literature, it is common practice to postulate that per-period vacancy posting costs are constant. Beaudry et al. (2018) have already partially abandoned this assumption and model a one-for-one relationship between wages and vacancy posting cost, $\chi = \chi_0 \cdot W$, implying that a 1% increase in wages always translates into a 1% increase in vacancy

²In line, Muehleemann and Strupler Leiser (2018) found a positive relationship between labor market tightness and pre-match hiring costs but no such association for post-match hiring costs.

posting costs. To allow for more flexibility, we will now deviate from their approach when formulating the functional form of vacancy posting cost.

Specifically, we work with the following formulation of vacancy posting cost per period:

$$(4) \quad \chi = \chi_0 \cdot W^{\phi_1} \cdot \theta_z^{\phi_2}$$

where $\chi_0 \geq 0, \phi_1 \geq 0$, and $\phi_2 \geq 0$. Equation (4) differs from the formulation in Beaudry et al. (2018) in two important aspects.³ On the one hand, we allow for the possibility that, due to intensified search efforts of firms, higher wages raise vacancy posting costs, but we do not fix the exponent of the wage rate to 1. On the other hand, we explicitly model that an increase in tightness not only prolongs the search duration, as given by Equation (3), but also raises vacancy posting costs per period. When labor markets tighten, firms will increase their recruiting intensity in terms of a higher search effort (e.g., by using more search channels) to improve matching efficiency (Davis, Faberman, and Haltiwanger 2013; Carrillo-Tudela, Gartner, and Kaas 2023).

Augmented Labor Demand Equation

To study the role of search frictions for firms' labor demand, we combine the labor demand equation, the job-creation curve, and our formulation of vacancy posting costs by inserting Equations (3) and (4) into Equation (1). After factoring out W , optimal labor demand of the representative firm is given by:

$$(5) \quad \begin{aligned} \ln L^* = & \frac{1}{1 - \alpha_1 - \alpha_2} \cdot \ln \left(\alpha_1^{1-\alpha_2} \cdot \alpha_2^{\alpha_2} \right) \\ & - \frac{1 - \alpha_2}{1 - \alpha_1 - \alpha_2} \cdot \ln W \\ & - \frac{1 - \alpha_2}{1 - \alpha_1 - \alpha_2} \cdot \ln (1 + (r + \delta) \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}) \\ & + \frac{1}{1 - \alpha_1 - \alpha_2} \cdot \ln (P \cdot A \cdot R^{-\alpha_2}) \end{aligned}$$

In the empirical part of the article, we will estimate a log-linearized version of Equation (5) to highlight the first-order effects of wages and labor market tightness (as a proxy for search frictions) on firms' labor demand. After log-linearization, the differenced equation becomes:

³The formulation of vacancy posting cost in Beaudry et al. (2018) represented a special case of our more general specification when $\phi_1 = 1$ and $\phi_2 = 0$.

$$\begin{aligned}
 (6) \quad \Delta \ln L^* = & -\frac{1-\alpha_2}{1-\alpha_1-\alpha_2} \cdot \frac{1+(r+\delta) \cdot \phi_1 \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}}{1+(r+\delta) \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}} \cdot \Delta \ln W \\
 & -\frac{1-\alpha_2}{1-\alpha_1-\alpha_2} \cdot \frac{(r+\delta) \cdot (\phi_2+\nu) \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}}{1+(r+\delta) \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}} \cdot \Delta \ln \theta_z \\
 & + \frac{1}{1-\alpha_1-\alpha_2} \cdot \Delta \ln (P \cdot A \cdot R^{-\alpha_2})
 \end{aligned}$$

To simplify the notation, we further rewrite the log-linear labor demand equation as

$$\begin{aligned}
 (7) \quad \Delta \ln L^* &= \beta_1 \cdot \Delta \ln W + \beta_2 \cdot \Delta \ln \theta + \Delta \varepsilon \\
 \text{with } \beta_1 &= -\frac{1-\alpha_2}{1-\alpha_1-\alpha_2} \cdot \frac{1+(r+\delta) \cdot \phi_1 \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}}{1+(r+\delta) \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}} < 0, \\
 \beta_2 &= -\frac{1-\alpha_2}{1-\alpha_1-\alpha_2} \cdot \frac{(r+\delta) \cdot (\phi_2+\nu) \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}}{1+(r+\delta) \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}} \leq 0, \text{ and } \Delta \varepsilon = \frac{1}{1-\alpha_1-\alpha_2} \cdot \\
 & \Delta \ln (P \cdot A \cdot R^{-\alpha_2}).
 \end{aligned}$$

Equation (7) provides a thorough framework to simultaneously study the relationship between wages, labor market tightness, and labor demand. To be precise, this specification not only postulates a negative wage effect on labor demand but also a negative tightness effect through search frictions. In the absence of vacancy posting costs, $\chi_0 = 0$, the specification collapses to the standard (unconditional) specification of labor demand with a wage effect only. Note that the derivation of Equation (7) is based on the steady-state assumption, which is why the specification is most adequately used to estimate medium- or long-run responses, which we will do in the empirical part of the article.

Magnitude of Hiring Costs

Interestingly, in our rich framework, the relative magnitude of the wage and the tightness effect allows us to determine the magnitude of hiring costs by specifying only a few model parameters. To achieve this, we use Equation (7) to divide the tightness effect by the wage effect on labor demand, which fundamentally simplifies to:

$$\begin{aligned}
 (8) \quad \frac{\beta_2}{\beta_1} &= \frac{\frac{1-\alpha_2}{1-\alpha_1-\alpha_2} \cdot \frac{(r+\delta) \cdot (\phi_2+\nu) \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}}{1+(r+\delta) \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}}}{\frac{1-\alpha_2}{1-\alpha_1-\alpha_2} \cdot \frac{1+(r+\delta) \cdot \phi_1 \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}}{1+(r+\delta) \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}}} \\
 &= \frac{(r+\delta) \cdot (\phi_2+\nu) \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}}{1+(r+\delta) \cdot \phi_1 \cdot \chi_0 \cdot W^{\phi_1-1} \cdot \theta_z^{\phi_2+\nu}}
 \end{aligned}$$

Recall that overall hiring cost C comprises a monetary dimension in terms of vacancy posting cost per period, χ , and a temporal dimension in terms of

search duration, $q(\theta_z)^{-1}$. Given our flexible formulation of vacancy posting cost (see Equation (4)), overall hiring cost then becomes:

$$(9) \quad C = \chi \cdot q(\theta_z)^{-1} = \chi \cdot \theta_z^\nu = \chi_0 \cdot W^{\phi_1} \cdot \theta_z^{\phi_2 + \nu}$$

Expanding Equation (8) by $\frac{W}{W}$ and combining with Equation (9) yields

$$(10) \quad \frac{\beta_2}{\beta_1} = \frac{(r + \delta) \cdot (\phi_2 + \nu) \cdot C}{W + (r + \delta) \cdot \phi_1 \cdot C}$$

Finally, we solve this expression for overall hiring costs as a fraction of wage payments:

$$(11) \quad \frac{C}{W} = \left((r + \delta) \cdot \left(\frac{\beta_1}{\beta_2} \cdot (\phi_2 + \nu) - \phi_1 \right) \right)^{-1}$$

In the empirical part of the article, we will use this relationship to gauge the magnitude of hiring costs as a fraction of annual wage payments.

Extended Model: Wage-Setting Curve

In Online Appendix A.2, we derive an extended version of our baseline model in which we, unlike Beaudry et al. (2018), additionally account for the process of wage formation by explicitly modeling the worker side of the labor market. In this extended version, we allow that higher labor market tightness may also raise wages by integrating a wage-setting-curve relationship. In the empirical part, we will show that the magnitude of the relationship between tightness and wages, β_3 , is rather small. Thus, our main conclusions will have little change when applying our empirical results to the extended model.

Search Externalities at the Aggregate Level

Both the baseline and the extended model highlight that not only higher wages but also higher labor market tightness reduce firms' labor demand. Unlike traditional models of labor demand, this framework implicitly incorporates search (or congestion) externalities from firms' labor demand decisions at the aggregate level, namely that an increase in employment in a certain firm complicates recruitment in other firms by intensifying search frictions (and vice versa). Given these externalities, aggregate changes in labor demand feature a self-attenuating feedback mechanism via labor market tightness.

The impact of a single firm's change in employment on labor market tightness is certainly negligible when the firm is small in relation to the overall size of the labor market. Even when the labor market is atomistic, however, the feedback mechanism becomes relevant when many firms alter their labor demand simultaneously (e.g., from responding to an increase in

a nationwide minimum wage). When firms act in concert, an aggregate decline in labor demand will reduce labor market tightness, which in turn will stimulate labor demand of individual firms. Ultimately, this self-dampening feedback cycle implies that the aggregate reduction in labor demand becomes less negative than the sum of firms' individual first-round responses due to reallocation of workers across firms.

Beveridge Curve

In the remainder of this section, we will derive the feedback mechanism formally by modeling the Beveridge curve. In our previous considerations, we derived optimal labor demand for a representative firm. In the following, we add the subscript i to differentiate between I firms. Thus, aggregate labor demand in a commuting zone z is given by: $L_z = \sum_{i=1}^I L_i$. At the aggregate level, the law of motion for employment implies that $\dot{L}_z = M_z - \delta \cdot L_z = U_z^\nu \cdot V_z^{1-\nu} - \delta \cdot L_z$. Again, we assume that the economy is in the steady state: $\dot{L}_z = 0$. Considering that regional population, $N_z = L_z + U_z$, is the sum of employed workers and unemployed persons, we insert the steady-state condition into the law of motion to arrive at the following version of the Beveridge curve:

$$(12) \quad \theta_z = \left(\frac{\delta \cdot L_z}{N_z - L_z} \right)^{\frac{1}{1-\nu}}$$

In logs, the relationship looks as follows:

$$(13) \quad \ln \theta_z = \frac{1}{1-\nu} \cdot \ln \delta + \frac{1}{1-\nu} \cdot \ln L_z - \frac{1}{1-\nu} \cdot \ln (N_z - L_z)$$

Under the assumption that population remains constant, $\dot{N}_z = 0$, the log-linearized version of Equation (13) is

$$(14) \quad \Delta \ln \theta_z = \beta_4 \cdot \Delta \ln L_z + \Delta \xi$$

where $\beta_4 = \frac{1}{1-\nu} \cdot \left(1 - \frac{\partial \ln U}{\partial \ln L}\right) > 0$, and ξ is the idiosyncratic error term.

Search Externalities and Reallocation Effects

As given by Equation (14), a first-round reduction in aggregate labor demand by 1% lowers labor market tightness by β_4 percent. This reduction in labor market tightness relieves some congestion in the hiring process and, by virtue of Equation (7), stimulates the demand for labor in all other firms by $\omega = \beta_4 \cdot \beta_2$ percent.⁴ This second-round increase in labor demand will, in turn, raise labor market tightness, which again translates into a third-

⁴Conventional models of labor demand rule out such a built-in feedback cycle by assuming zero vacancy posting cost, $\chi_0 = 0$. In this case, tightness exerts no effect on labor demand: $\beta_2 = 0$.

round reduction in labor demand by ω^2 (and so forth). Overall, aggregate changes in labor demand (regardless of their trigger) generate a self-dampening feedback cycle to the extent that aggregate labor demand creates congestion in the hiring process through increased tightness. Using a geometric series, we can derive that only the fraction $\frac{1}{1-\omega}$ of the first-round response in labor demand survives the self-attenuating feedback cycle.⁵

In terms of the wage effect, knowledge about the strength of the feedback mechanism, ω , allows us to derive the aggregate own-wage elasticity of labor demand

$$(15) \quad \beta_1^{agg} = \frac{\beta_1}{1-\omega} = \frac{\beta_1}{1-(\beta_4 \cdot \beta_2)}$$

which, by accounting for search externalities, captures the ultimate response of aggregate labor demand to aggregate wage changes (Beaudry et al. 2018). When, according to Equations (7) and (14), ω is negative, the aggregate own-wage elasticity of labor demand β_1^{agg} becomes less negative than the own-wage elasticity of labor demand β_1 of an individual firm due to reallocation effects.

Empirical Design

In the next step, we describe our empirical design to trace out the relationships between wages, labor market tightness, and labor demand using plausibly exogenous variation from shift-share instruments. For further details, we direct the reader to Online Appendix B.

Empirical Model

To estimate the causal effects of wages and labor market tightness, we specify the following empirical version of our augmented log-linear labor demand Equation (7) in first differences

$$(16) \quad \Delta \ln L_{it} = \beta_0 + \beta_1 \cdot \Delta \ln W_{it} + \beta_2 \cdot \Delta \ln \theta_{it} + \zeta_{zt} + \Delta \tilde{\varepsilon}_{it}$$

where L_{it} denotes employment in firm i in year t , W_{it} is the firm's average wage, θ_{it} captures firm-specific labor market tightness, ζ_{zt} represents commuting-zone-by-year fixed effects, and $\tilde{\varepsilon}_{it}$ is the idiosyncratic error term (excluding the fixed effects). In the baseline specification, we estimate our empirical model (16) in two-year differences. These biennial changes allow adjustments to take two years to materialize, identifying firms' responses in the medium run.

Labor market tightness is typically measured at the level of regional labor markets. However, regional labor market tightness may not be a precise

⁵Note that the geometric series converges only when $|\omega| = |\beta_4 \cdot \beta_2| < 1$.

measure for firms' demand if a firm recruits only from specific occupational labor markets within a region. To account for the occupational demands of individual firms, we develop a measure of firm-specific labor market tightness

$$(17) \quad \theta_{it} = \sum_{o=1}^O \frac{L_{oit}}{L_{it}} \cdot \tilde{\theta}_{ozt} = \sum_{o=1}^O \frac{L_{oit}}{L_{it}} \cdot \frac{V_{ozt}}{U_{ozt}}$$

where $\frac{V_{ozt}}{U_{ozt}}$ is the ratio between vacancies in occupation o , in commuting zone z , in year t and the number of job seekers in the same occupation-by-zone-by-year cell. We weight these occupational measures of labor market tightness in a firm's commuting zone by the respective shares of these occupations in each firm's overall employment, $\frac{L_{oit}}{L_{it}}$.

Threats to the Identification

While differencing eliminates unobserved time-invariant heterogeneity capturing all permanent differences between firms (including the industry, the location, or firms' permanent growth potential), we still require exogeneity of the differenced independent variables to establish causality. From Equation (6), we know that changes in the product price, total factor productivity, and the rental rate of capital enter the differenced error term. Hence, we must ensure that our variation in wages or tightness does not stem from changes in these omitted labor demand variables at the respective firm.

Specifically, the threats of identification are twofold. First, endogeneity of the wage and tightness variable may arise from omitted changes in labor demand variables. Ideally, variation in wages should represent movements along the labor demand curve rather than shifts of the curve itself. Given the positive relationship between wages and labor supply, uncontrolled shifts of the labor demand curve will result in an upward bias. In addition, uncontrolled changes in labor demand (e.g., productivity shocks), which positively affect tightness through firms' vacancy creation, will lead to an upward-biased estimate for tightness.

Second, endogeneity of the tightness variable may also arise from reverse causality. Specifically, we are interested in the effects of labor market tightness on firms' employment. At the same time, however, Equation (14) implies that any change in employment directly impacts labor market tightness. As an increase in labor demand will raise tightness, the feedback mechanism leads to an upward bias, even if the single firm may just have a small influence on the respective tightness at the market level.

Identification Strategy

Since a naive OLS estimation of Equation (7) may provide biased results, we estimate our model based on variation from instrumental variables using

two-stage least squares (2SLS) estimation. We propose three new shift-share instruments in the tradition of Bartik (1993). Bartik instruments exploit the inner product structure of endogenous variables to deliver plausibly exogenous variation at the regional level (Goldsmith-Pinkham, Sorkin, and Swift 2020). However, for the purpose of analyzing reallocation effects of workers across employers, we do not seek to identify employment effects at the level of regions but for individual firms. Thus, we develop novel Bartik instruments that provide variation at the firm level.⁶ In particular, we take advantage of the fact that firms differ in the (past) occupational composition of their workforce and, thus, are differently exposed to national shocks.

Our Bartik instrument that is meant to exogenously predict wage changes at the firm level, Z^W , looks as follows:

$$(18) \quad Z_{it}^W = \sum_{o=1}^O \frac{L_{io\tau}}{L_{i\tau}} \cdot \Delta \ln W_{ot}$$

We analogously define two separate Bartik instruments for vacancies and job seekers, Z^V and Z^U , which are meant to generate exogenous variation in firm-specific tightness:

$$(19) \quad Z_{it}^V = \sum_{o=1}^O \frac{L_{io\tau}}{L_{i\tau}} \cdot \Delta \ln V_{ot}$$

$$(20) \quad Z_{it}^U = \sum_{o=1}^O \frac{L_{io\tau}}{L_{i\tau}} \cdot \Delta \ln U_{ot}$$

Specifically, our three shift-share instruments combine the occupational mix of employment within firms at an early period τ with occupation-specific growth rates of average wages, vacancies, or job seekers at the national level. To ensure exogeneity, the instruments depart from observed growth in the instrumented variables in two dimensions. First, by fixing the occupational shares in the past, we minimize that the predetermined shares correlate with differences in the contemporaneous error term.⁷ Second, by relying on nationwide rather than firm-specific growth rates, we mitigate that these shifts are driven by firm-specific labor demand shocks in the differenced error term. By virtue of the shift-share design, the exclusion restriction of our three Bartik-style instruments Z^W , Z^V , and Z^U is fulfilled when either the national growth rates (i.e., the shifts) or the predetermined firm-specific occupational composition (i.e., the shares) are uncorrelated

⁶We are aware of only two other contributions that construct shift-share instruments at the level of firms, namely in the context of offshoring (Hummels, Jørgensen, Munch, and Xiang 2014) and technological change (Aghion, Antonin, Bunel, and Jaravel 2022).

⁷In particular, the predetermination of the shares also rules out endogeneity from firms that substitute toward cheaper or slacker occupations.

with the differenced error term (Goldsmith-Pinkham et al. 2020; Borusyak, Hull, and Jaravel 2022).

Although some of the variation in wage growth may stem from the exogenous first-time introduction of the minimum wage in 2015, it is difficult to show that the nationwide shifts in vacancies and job seekers are solely driven by labor supply shocks or firm-specific labor demand shocks. When exposure to labor demand shocks is correlated between units (e.g., a common technology shock), the use of nationwide shifts in the Bartik instrument no longer protects against omitted variable bias and reverse causality.

In light of our setting, we instead argue that our identification strategy mainly rests on the exogeneity of the predetermined shares. Thus, we build on an exposure design in which firms are differently affected by (non-exogenous) shocks based on their occupational employment mix from the past. As the shares are equilibrium outcomes, they may be codetermined with the level of the outcome variable. However, the differencing in Equation (16) eliminates unobserved level differences in labor demand between firms. Thus, our identifying assumption only requires that the predetermined shares must be exogenous to changes (rather than levels) in the error term (i.e., uncontrolled labor demand shocks). In the empirical part of the article, we will follow the guidelines of Borusyak, Hull, and Jaravel (2025) to prove the plausibility of our share-based identification strategy.

Data

In the following section, we briefly describe our main data sets, the construction of our tightness measure, and the operationalization of our shift-share instrumental variables. Additional explanations are given in Online Appendix C.

Integrated Employment Biographies

The Integrated Employment Biographies (IEB) comprise the administrative labor market records of Germany (Müller and Wolter 2020). From the IEB, we use the universe of employment records of all workers subject to social security contributions, which are collected from employers in Germany as part of the mandatory reporting requirement.⁸ The IEB provides day-to-day information on workers' employment histories, such as workers' establishment, daily gross wages (which we impute above the censoring limit), type of contract, place of work, as well as an indicator of whether workers have a full- or part-time contract. Importantly, the IEB data also offer exceptionally

⁸The data exclude only self-employed persons, civil servants, and family workers because these groups are not obliged to pay social security contributions.

rich information on workers' 5-digit occupation, distinguishing between a total of $O = 1,286$ occupational categories.⁹

For June 30 of the years 2012–2019, we construct a panel data set by calculating the number of workers and average daily wages for each firm.¹⁰ The term “establishment” (or “firm”) comprises all plants of a company that share the same economic activity within a municipality.¹¹ For lack of information on individual working hours, we follow standard practice and restrict our baseline analysis to full-time workers in regular employment (who are supposed to work a similar number of hours). Throughout the study, we exclude apprentices and people in partial retirement schemes. In total, our final data set covers the near-universe of firms in Germany and contains a total of 21,698,311 firm-year observations from 4,207,641 firms.

Labor Market Tightness

We define labor markets as combinations of 5-digit occupations and commuting zones. In terms of occupations, we employ the 5-digit classification for two reasons. First, the leading four digits differentiate between 700 types of occupations in the highest available level of detail. Second, the fifth digit further delivers valuable information on the level of skill requirement, namely whether workers are helpers, professionals, specialists, or experts. It is highly important to distinguish between requirement levels since tasks with different levels of complexity plausibly define segregated labor markets even if the underlying 4-digit occupation is identical. In terms of regions, we employ the graph-theoretical method from Kropp and Schwengler (2016) and merge 401 administrative districts (“Kreise”) to functional labor market regions that reflect commuting patterns (see Appendix Figure C1). Taken together, our baseline labor markets constitute combinations of $O = 1,286$ occupations and $Z = 51$ commuting zones.

We gather administrative data on posted vacancies and job seekers from the Federal Employment Agency (FEA) to construct our measure of firm-specific labor market tightness. For each June 30 between 2012 and 2019, we draw official statistics on the stock of registered vacancies (FEA 2019), including the targeted 5-digit occupation and commuting zone (in terms of workplace). In Germany, there is no obligation for firms to register vacancies with the FEA. To quantify the overall stock of registered plus unregistered vacancies for each labor market and year, we divide the number of registered vacancies by the yearly share of registered vacancies from

⁹In particular, we utilize information on the German Classification of Occupations (KldB) from the year 2010. The four leading digits describe the type of occupation, whereas the fifth digit designates the level of skill requirement (helper, professional, specialist, or expert).

¹⁰Although data availability would support construction of our detailed measures of labor market tightness from 2010 onward, we choose 2012 as the starting year for our period of analysis since there was a structural break in 2011/12 in the occupation variable in the IEB data.

¹¹Throughout this study, we use the terms “establishment” and “firm” interchangeably.

the IAB Job Vacancy Survey (Bossler et al. 2020). The IAB Job Vacancy Survey (IAB-JVS) is a representative firm survey with a focus on recruitment behavior and, in particular, asks firms about their number of registered and unregistered vacancies. When dividing by the yearly shares of registered vacancies, we differentiate between three levels of skill requirement: occupations for helpers, for professionals, and for specialists along with experts.¹²

For job seekers, it is mandatory to register as unemployed with the FEA to be eligible for benefits from unemployment insurance or social assistance. For the same labor market and years, we extract official information on the number of job seekers (FEA 2018), namely registered unemployed plus non-unemployed workers searching for a job via the Federal Employment Agency.¹³ Upon registration, job-seeking individuals must submit their targeted 5-digit occupation to the FEA. For each labor market and year, we divide the overall stock of registered plus unregistered vacancies by the stock of job seekers. Next, we apply Equation (17) and weight these ratios with contemporaneous shares of 5-digit occupations in firms' overall employment from the IEB to arrive at our measure of firm-specific labor market tightness.

Shift-Share Instruments

In a final step, we build our firm-level shift-share instruments from Equations (18), (19), and (20). To this end, we interact the biennial national changes in average wages, stock of vacancies, and stock of job seekers per occupation with IEB information on firms' occupational shares in their employment from the past. When choosing the base year of these shares, we face a trade-off between maximizing the time lag to the estimation period (i.e., 2012–2019) and minimizing structural breaks in the data over time. Because of a major redesign of the IEB data, we calculate the predetermined employment shares only from 1999 onward.¹⁴ Hence, in most cases, the base year refers to 1999 (48.0%) or, alternatively, the year of birth for firms that entered the labor market at a later stage (0.6–3.3% per year from 2000 onward).

¹²From the IAB Job Vacancy Survey, we calculate the following notification shares for vacancies, averaged over 2012–2019: helpers (46.1%), professionals (45.6%), specialists and experts (31.1%).

¹³Abraham et al. (2020) found that the number of effective job searchers features a higher explanatory power in the matching function than the mere stock of unemployed persons.

¹⁴In principle, IEB information is available from 1975 (West Germany) and 1993 (East Germany) onward. We refrain from analyzing information from before 1999, however, because information on marginal employment has not yet been available. For the years 1999–2011, we rely on available crosswalks to translate the information on the occupational classification from 1988 (KldB-1988) into time-consistent information on the occupational classification from 2010 (KldB-2010), which is available in the IEB from 2012 onward.

Results: Labor Market Tightness

Before estimating our augmented labor demand equation, we inspect our measure of (firm-specific) labor market tightness and assess its performance as a main determinant for pre-match hiring costs. In this regard, Online Appendix D delivers additional evidence.

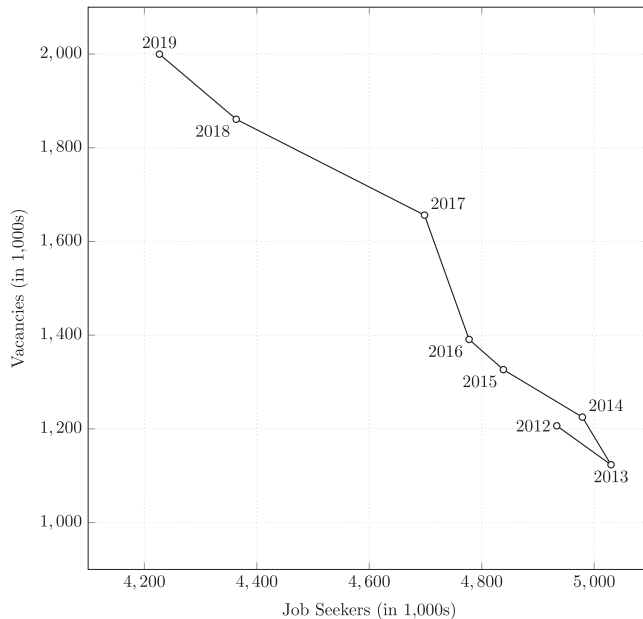
The German Economy

After having risen steadily for several decades, Germany's unemployment rate reached a peak of 13% in the mid-2000s. During that time, the muted economic environment also deterred many firms from posting vacancies. Thus, labor market tightness in Germany reached an all-time low in 2004 (see Appendix Figure D1). Since then, the German labor market has undergone a remarkable transformation, accompanied by significant employment growth. Dustmann, Fitzenberger, Schönberg, and Spitz-Oener (2014) attributed this reversal to the flexibility and decentralization of the wage-setting process resulting in higher labor demand from lower real wages. In addition, a comprehensive reform of German labor market institutions in the years 2003–2005 (the so-called Hartz laws) contributed to the labor market upswing (Krause and Uhlig 2012; Hochmuth, Kohlbrecher, Merkl, and Gartner 2021). Among others, these laws restructured the Federal Employment Agency and reduced the generosity of unemployment benefits to increase workers' incentive to accept jobs, thus further weakening workers' bargaining position. A number of studies demonstrate that the Hartz reforms also came along with an increased matching efficiency (Fahr and Sunde 2009; Klinger and Rothe 2012; Launov and Wälde 2016). As a side effect of the favorable employment growth, labor market tightness started to increase (Burda and Seele 2020). The economic prosperity continued in the following decade. Simultaneously, demographic change led to a decline in the number of unemployed, especially in East Germany (Schneider and Rinne 2019). As a result, the increase in labor market tightness accelerated during the 2010s.

Beveridge Curve

Figure 1 displays the Beveridge curve for Germany for our period of analysis. Between 2012 and 2013, labor market tightness decreased slightly during the sovereign debt crisis in the Euro area. From 2013 to 2019, we observe a sharp increase in the number of vacancies by 800,000 and a decrease in the number of job seekers by a similar magnitude. Between 2012 and 2019, the economy-wide ratio of vacancies to job seekers rose from 0.24 to 0.47: We report four job seekers per vacancy in 2012, but there were only two job seekers per vacancy in 2019, implying a doubling in labor market tightness.

Figure 1. Beveridge Curve



Sources: Official Statistics from the Federal Employment Agency, IAB Job Vacancy Survey, 2012–2019.

Notes: The figure shows the Beveridge curve for Germany between 2012 and 2019. The numbers of registered vacancies and job seekers stem from notifications to the German Federal Employment Agency. We divide the stock of registered vacancies by the yearly share of registered vacancies per requirement level from the IAB Job Vacancy Survey to account for unregistered vacancies.

Our period of analysis coincides with a significant and long-lasting phase of prosperity of the German economy, which came along with a rapid employment expansion, rising from 41.6 million employees in 2012 to 45.4 million employees in 2019. Survey evidence shows that the observed tightening was accompanied by a markedly higher share of firms that face labor shortages, implying that firms' employment could have grown even more if tightness had stayed constant.

Labor Market Tightness, Wages, and Hiring Costs

In the theory section, labor market tightness is hypothesized to exert a negative effect on the demand for labor through increased vacancy posting costs. In a next step, we use the IAB Job Vacancy Survey to provide empirical support for this channel by estimating the effect of labor market tightness on pre-match hiring costs and several other recruitment indicators.

In repeated cross-sections, the IAB-JVS asks firms about their most recent successful hiring process. The survey includes information on the following recruitment indicators: direct pre-match hiring costs (in Euros), search effort (in working hours), the number of applicants, the number of search channels, and search duration (in days). In addition, we combine the survey

information on search effort (in working hours) with IEB information on the respective firm's average wage rate to measure indirect pre-match hiring costs (in Euros). Based on the firm's location, targeted 5-digit occupation, and year, we enrich our survey-based recruitment indicators with the tightness of the respective labor market.

Initially, we focus on the raw correlation between labor market tightness and pre-match hiring costs. In a more sophisticated specification, we estimate the following log-linear version of our formulation of vacancy posting cost of Equation (4),

$$(21) \quad \ln \chi_{it} = \phi_0 + \phi_1 \cdot \ln W_{it} + \phi_2 \cdot \tilde{\theta}_{ozt} + \zeta_o + e_{iozt}$$

where χ_{it} represents either direct, indirect, or overall pre-match hiring costs of the last successful hire in firm i and year t , W_{it} is the hourly entry wage of the new hire, $\tilde{\theta}_{ozt}$ is the ratio of vacancies to job seekers in the new hire's occupation o and the firm's commuting zone z in year t , ζ_o are occupation fixed effects, and e_{iozt} is the idiosyncratic error term.

Table 1, panel (a), shows the relationship between labor market tightness and pre-match hiring costs. The simple regressions in columns (1), (3), and (5) highlight that a positive raw correlation occurs between labor market tightness and direct, indirect, and overall pre-match hiring costs. In columns (2), (4), and (6), we present the results for our more sophisticated specification with log entry wages and 5-digit occupation fixed effects as controls. The regressions show that a doubling in labor market tightness (i.e., an increase by 100%) raises direct pre-match hiring costs by 12.7%, indirect pre-match hiring costs by 5.2%, and overall pre-match hiring costs by 14.8% (i.e., $\phi_2 > 0$). Conditional on the occupation, higher wages also come along with higher pre-match hiring costs (i.e., $\phi_1 > 0$). This positive wage coefficient implies that firms intensify their search (or screening) efforts when they offer higher wages. Specifically, an increase in the entry wage by 100% raises direct, indirect, and overall pre-match hiring costs by 52.4, 40.2, and 62.4%, respectively.

Panel (b) of Table 1 shows the relationship between labor market tightness and three recruitment indicators that mediate the observed effect on pre-match hiring costs: the number of applicants, the number of search channels, and search duration. All effects feature the expected sign. More precisely, a doubling in labor market tightness reduces the number of applicants by 14.4%, raises the number of search channels by 5.8%, and prolongs the search duration by 6.0%.

By and large, the documented relationships between labor market tightness, wages, and pre-match hiring costs provide an empirical foundation for our formulation of hiring cost in Equation (4). Specifically, the regressions substantiate the theoretical mechanism that higher (firm-specific) labor market tightness raises hiring costs of firms, and thus, may manifest in a negative causal effect on firms' demand for labor.

Table 1. Relationship between Labor Market Tightness, Wages, and Recruitment Indicators

(a) Effect on pre-match hiring cost					
	(1) Log direct hiring costs	(2) Log direct hiring costs	(3) Log indirect hiring costs	(4) Log indirect hiring costs	(5) Log overall hiring costs
Log labor market tightness	0.109*** (0.011)	0.127*** (0.019)	0.051*** (0.006)	0.052*** (0.010)	0.148*** (0.014)
Log hourly entry wage		0.524*** (0.032)		0.402*** (0.017)	0.624*** (0.024)
5-digit occupation fixed effects	No	Yes	No	Yes	No
Observations	13,733	12,826	31,999	29,749	26,918
					25,237
(b) Effect on recruitment indicators					
	(1) Log applicants	(2) Log applicants	(3) Log search channels	(4) Log search channels	(5) Log search duration
Log labor market tightness	-0.123*** (0.005)	-0.144*** (0.010)	0.053*** (0.002)	0.058*** (0.005)	0.161*** (0.005)
Log hourly wage rate		0.086*** (0.014)		0.020*** (0.007)	0.095*** (0.015)
5-digit occupation fixed effects	No	Yes	No	Yes	No
Observations	48,780	33,581	57,832	39,372	47,317
					33,320

Sources: Official Statistics from the Federal Employment Agency, IAB Job Vacancy Survey, 2012–2019.

Notes: The table shows the effect of labor market tightness and hourly entry wages on pre-match hiring costs as well as several recruitment indicators. The three recruitment indicators are the number of applicants, the number of search channels, and search duration (in days). We differentiate between direct, indirect, and overall pre-match hiring costs. Pre-match hiring costs and the hourly entry wage were deflated using 2015 as the base year. Labor markets are pairs of 5-digit KldB occupations and commuting zones. Regressions are weighted using survey weights. KldB = German Classification of Occupations.

* = $p < 0.10$; ** = $p < 0.05$; *** = $p < 0.01$.

Results: Labor Demand Effects

In the following section, we quantify the extent to which the tightening of labor markets reduces firms' labor demand (and ultimately employment), while simultaneously determining the own-wage elasticity of labor demand.

Baseline Results

Table 2 displays the baseline estimates, including potentially endogenous OLS estimates as well as instrumental variable estimates from 2SLS. Column (1) presents results from a naive OLS estimation of Equation (16). While the own-wage elasticity of labor demand is negative, albeit small, the elasticity with respect to tightness turns out to be positive, unlike predicted by theory. As pointed out above, however, the OLS estimates may feature an upward bias.

To address the bias in either case, we use our Bartik-style instruments in Equations (18), (19), and (20) to isolate plausibly exogenous variation in wages and labor market tightness in a second, third, and fourth specification. In column (2), we solely estimate the effect of wages on labor demand, using the wage instrument (18). As expected in comparison to OLS, the wage elasticity of labor demand turns out to be more negative, implying that firms lower employment by 0.73% when wages increase by 1%. Building on the instruments for vacancies (19) and job seekers (20), column (3) displays the IV effect of tightness on employment without conditioning on wages. In contrast to OLS, the elasticity turns negative, indicating that an increase in tightness by 1% reduces firms' employment *ceteris paribus* by 0.05% in the medium run after two years.

Finally, column (4) displays the effects of wages and tightness from a joint IV model, in which both variables exert an additive impact on firms' labor demand. We refer to this model as our baseline specification, which is described by Equation (16). In our baseline estimation, the own-wage elasticity of labor demand is -0.72 . Since our shift-share design rigorously addresses upward bias, this elasticity is at the lower end of the values found in the international and German literature (Lichter et al. 2015; Popp 2023). The elasticity of labor demand with respect to tightness is -0.05 , implying that the observed doubling in tightness between 2012 and 2019 (i.e., an increase by 100%) reduced firms' employment *ceteris paribus* by 5%, holding all other things equal. The effects are significant at the 1% level. Interestingly, both elasticities remain largely unchanged compared to the separate regressions in columns (2) and (3). On the one hand, this finding highlights that the instruments for wages and tightness do not interact with each other. On the other hand, by comparing column (4) with column (3), we can rule out that tightness substantially affects labor demand through

Table 2. Effects of Wages and Labor Market Tightness on Employment

	(1) ΔLog <i>employment</i>	(2) ΔLog <i>employment</i>	(3) ΔLog <i>employment</i>	(4) ΔLog <i>employment</i>
$\Delta \text{Log wage}$	-0.134*** (0.006)	-0.731*** (0.053)		-0.719*** (0.060)
$\Delta \text{Log tightness}$	0.050*** (0.005)		-0.053*** (0.006)	-0.050*** (0.005)
Fixed effects	Year $\times$ CZ	Year $\times$ CZ	Year $\times$ CZ	Year $\times$ CZ
Instruments	None	Z^W	Z^V, Z^U	Z^W, Z^V, Z^U
Observations	7,997,029	7,997,029	7,997,029	7,997,029
Clusters	51	51	51	51
$F: \Delta \text{Log wage}$		395		428
$F: \Delta \text{Log tightness}$			1,197	800

Sources: Integrated Employment Biographies, Official Statistics from the Federal Employment Agency, IAB Job Vacancy Survey, 1999–2019.

Notes: The table displays OLS and IV regressions of differences in log employment (of regular full-time workers) per firm on differences in the log of average daily wages and the log of firm-specific labor market tightness. The instrumental variables refer to shift-share instruments of biennial national changes in occupations weighted by past occupational employment in the respective firm. The lag difference is two years. Labor markets are combinations of 5-digit KldB occupations and commuting zones. Standard errors (in parentheses) are clustered at the commuting-zone level. CZ = commuting zone; F = F -statistics of excluded instruments; IV = instrumental variable; KldB = German Classification of Occupations; OLS = ordinary least squares; U = job seekers; V = vacancies; W = average daily wages; Z = shift-share instrument.

* = $p < 0.10$; ** = $p < 0.05$; *** = $p < 0.01$.

changes in wages because controlling for the wage channel hardly alters the tightness effect.¹⁵

Empirical Checks to the Identification Strategy

Next, we perform several empirical checks to assess the plausibility of our shift-share design. In the following, we will briefly discuss the main findings of these checks. For further details, see Online Appendix E.

First-Stage and Reduced-Form Effects

First-stage regressions show that our shift-share instruments are good predictors for changes in wages and tightness at the firm level. All three first-stage estimates show the expected sign and are significantly different from zero. The F -statistics for the average wage and firm-specific tightness are 428 and 800, respectively. In reduced-form regressions of the outcome

¹⁵The similarity of the tightness effect between column (3) and column (4) implies that, in Equation (A.33) in Online Appendix A, the relative effect of labor market tightness on wages, β_3 , is close to zero. In this case, the wage-setting-curve relationship is rather weak and, thus, any quantitative conclusions through the lens of our extended model resemble those from our baseline model.

variable on the shift-share instruments, each instrument shows the expected signs implied by the baseline IV estimates.

Examining the Plausibility of Exogenous-Share Assumption

In view of our setting, we argue that our identification strategy mainly rests on the exogeneity of the predetermined shares. Consequently, our identifying assumption requires that the predetermined shares must be exogenous to changes (rather than levels) in uncontrolled labor demand variables. To prove the plausibility of our exogenous-share assumption, we follow the guidelines of Borusyak et al. (2025) and perform five important checks.

First, we support the lagging of our shares by inspecting the dynamic lag structure of changes in firms' employment. The predictive power of the lags quickly decays after a few years, whereas our lagging of shares ensures an average interval of 13.0 years. Thus, the quick decay highlights that past shares may hardly correlate with the differenced contemporaneous error term. Second, we prove the robustness of our estimates by controlling for additional labor demand variables at the industry and firm level. Specifically, the coefficients are hardly altered when conditioning on industry-by-year fixed effects, firms' capital stock, and firms' productivity.

Third, we follow Goldsmith-Pinkham et al. (2020) and decompose the Bartik estimates into Rotemberg weights and just-identified IV estimates for each share-year combination. Favorably, the quantitative magnitude of negative Rotemberg weights is relatively small, which allows for a local average treatment effect (LATE) interpretation of our estimates. At the same time, the distribution of Rotemberg weights across occupations is highly skewed. Fourth, we focus on the occupations with the highest Rotemberg weights, to which the Bartik estimates are most sensitive, and show that the cross-sectional correlation between these occupational employment shares and the level of labor demand variables is rather weak. Thus, any correlation between predetermined shares and changes in labor demand variables in the far-off future is supposed to be even smaller, bolstering our exogenous-share assumption. Fifth, we show that there is heterogeneity in our just-identified IV estimates. Using the limited information maximum likelihood (LIML) regression technique, we rule out that the estimation suffers from a potential many-weak instrument bias.

Examining the Plausibility of Exogenous-Shift Assumption

By virtue of our shift-share design, exogeneity may, in principle, also arise when the nationwide shifts are uncorrelated with the differenced error term. This assumption may be partly justifiable for the wage instrument since the majority of Rotemberg weights refer to the years 2014/15 when Germany first introduced a nationwide minimum wage. A way to strengthen the exogenous-shift assumption is to adjust the national averages or totals by excluding the contribution of the focal (and similar) firms. When

constructing nationwide shifts, leaving out the respective firm's commuting zone or industry, it is comforting to see that our results hardly change.

Further Analyses

In the following, we perform further robustness checks and heterogeneity analyses. Figure 2 visualizes the corresponding results, and the respective regression tables are assembled in Appendices F and G.

Sensitivity

As elaborated earlier, we specify our empirical model in two-year differences to estimate medium- or long-run elasticities. With one-year differences, the wage and tightness effects turn out less negative, indicating that labor demand responses do not fully materialize in the short run due to adjustment costs.¹⁶ However, the elasticity barely increases for three-year differences.

In terms of model specification and measurement, the results turn out highly robust when including year fixed effects only, when relying on median wages, or when using registered vacancies only. We also test the sensitivity regarding our delineation of labor markets into pairs of 1,286 5-digit occupations and 51 commuting zones. First, we combine 3-digit occupations with their requirement level (5th digit) to form 438 “detailed 3-digit occupations.” Second, we construct a novel flow-based measure of labor market tightness that accounts for vacancies and job seekers in related occupations (see Online Appendix H). Third, in terms of regions, we rely on the administrative delineation of 401 districts (“Kreise”). In each of the three robustness checks, the results closely align with our baseline effects, buttressing that our findings are not driven by a particular labor market definition.

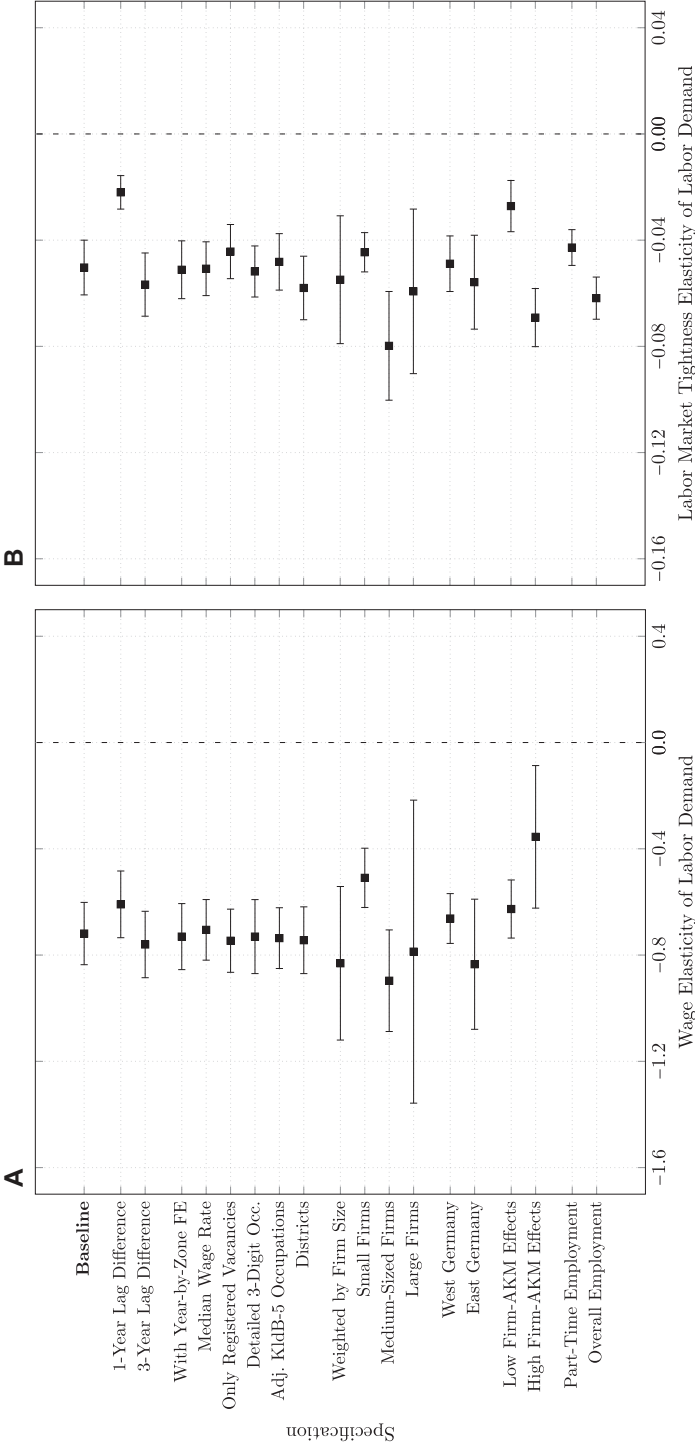
Heterogeneity

When weighting observations by employment to assign larger firms more importance, the results remain fairly robust. In comparison to our baseline, we observe somewhat smaller effects for small firms, somewhat larger effects for medium-sized firms, and similar effects for large firms.

The wage effect turns out slightly more negative in East Germany than in West Germany, while the tightness effects exhibit similar magnitudes. In line with rent sharing, the own-wage elasticity of labor demand for firms with high productivity, as indicated by AKM (Abowd, Kramarz, and Margolis 1999) firm fixed effects above the median, turns out to be less negative than for low-productive firms (i.e., with AKM firm fixed effects below the median). At the same time, the negative effect of labor market tightness is

¹⁶In Appendix F, we augment the specification in one-year differences with additional first lags of our differenced explanatory variables (Jaeger, Ruist, and Stuhler 2018). Both the main and the lagged effects feature negative signs, underlining that the effects accumulate over time.

Figure 2. Sensitivity and Heterogeneity of Labor Demand Effects



Sources: Integrated Employment Biographies (IEB), Official Statistics from the Federal Employment Agency, IAB Job Vacancy Survey, 1999–2019.

Notes: The figure illustrates estimated labor demand elasticities for a variety of specifications. The baseline IV estimation regresses differences in log employment (of regular full-time workers) per firm on differences in the log of average daily wages and the log of firm-specific labor market tightness. The instrumental variables (IV) for average wages, vacancies, and job seekers refer to shift-share instruments of biennial national changes in occupations weighted by past occupational employment in the respective firm. The baseline regression refers to regular full-time employment, and the lag difference is two years. Baseline labor markets are combinations of 5-digit KdIB occupations and commuting zones. Each point estimate features a 95% confidence interval. For lack of information on hours worked in the IEB, we can estimate wage effects only for the group of regular full-time workers. AKM = Abowd-Kramarz-Margolis. FE = fixed effects.

nearly three times smaller for low-productivity firms, reflecting that labor shortage poses a more severe problem to highly productive firms, which tend to grow more. Thus, the AKM heterogeneity suggests that the rise in tightness restricts additional employment growth rather than forcing firms to shrink.

For lack of data on hourly wages, our baseline analysis is restricted to daily wages of full-time workers (who are supposed to work a similar number of hours). Next, we discard the wage variable to inspect whether our tightness effect also holds for other groups of workers. In fact, we observe significantly negative tightness effects on part-time workers and all workers that closely resemble the effect on full-time workers (-0.05).

Wage and Skill Concessions

Firms facing higher labor market tightness do not necessarily have to settle for lower employment levels. Instead, these firms could still manage to retain or expand their workforce by increasing their recruitment intensity or, put differently, by making concessions. In this regard, we empirically address the conjecture that firms pay higher wages or lower their hiring standards to maintain their employment (growth) in tight labor markets.

In Table 3, we regress the average wage level of firms and the fraction of unskilled workers on our measure of labor market tightness, building on the same instrumental variable approach as in our baseline analysis. In column (1), the doubling in tightness raises average wages of full-time workers in a firm by almost 0.4%. While the wage response is significantly positive, the magnitude of this effect is fairly small, namely just about a fifth of the negative employment response.¹⁷ However, firms' positive but small wage response is in line with the empirical literature on the wage curve for Germany, which relates wages to the unemployment rate (Bellmann and Blien 2001; Baltagi, Blien, and Wolf 2009). Regarding skill demand, we also observe only a limited extent of concessions. Starting from an average share of 6.8%, our results in column (2) imply that the doubling in tightness raised the share of unskilled workers in firms' employment by only 0.4 percentage points. Overall, the estimates suggest that the extent of firms' wage and skill concessions was relatively small in practice, providing an explanation for the markedly negative effect of labor market tightness on employment.

Magnitude of Pre-Match Hiring Costs

As given by Equation (11), our rich theoretical framework allows us to quantify the magnitude of pre-match hiring costs (as a fraction of annual wage

¹⁷In line with this observation, when discarding the wage variable in column (3) of Table 2, the tightness effect hardly changes compared to the baseline in column (4), reflecting the relatively small extent of wage concessions.

Table 3. Effects of Labor Market Tightness on Wages and Share of Unskilled Workers

	(1) $\Delta \text{ Log wage}$	(2) $\Delta \text{ Share of unskilled workers}$
$\Delta \text{ Log wage}$		−0.009 (0.007)
$\Delta \text{ Log tightness}$	0.004** (0.002)	0.004*** (0.000)
Fixed effects	Year $\times$ CZ	Year $\times$ CZ
Instruments	Z^V, Z^U	Z^W, Z^V, Z^U
Observations	7,997,029	7,861,939
Clusters	51	51
F : $\Delta \text{ Log wage}$		359
F : $\Delta \text{ Log tightness}$	1,197	820

Sources: Integrated Employment Biographies, Official Statistics from the Federal Employment Agency, IAB Job Vacancy Survey, 1999–2019.

Notes: The table displays IV regressions of differences in average wages per firm and the share of unskilled workers per firm on differences in the log of average hourly wages and the log of labor market tightness. The instrumental variables (IV) refer to shift-share instruments of biennial national changes in occupations weighted by past occupational employment in the respective firm. The lag difference is two years. Labor markets are combinations of 5-digit KldB occupations and commuting zones. Full-time employment includes regular full-time workers, whereas part-time employment encompasses regular part-time and marginal part-time workers. Unskilled workers have neither completed vocational education nor acquired a university degree. Standard errors (in parentheses) are clustered at the commuting-zone level. F = F -statistics of excluded instruments; KldB = German Classification of Occupations; U = job seekers; V = vacancies; W = average hourly wages; Z = shift-share instrument.

* = $p < 0.10$; ** = $p < 0.05$; *** = $p < 0.01$.

payments) by calibrating only few parameters. In this formula, the ratio of the wage elasticity of labor demand to the elasticity of labor demand with respect to tightness, $\frac{\beta_1}{\beta_2}$, is the key determinant. Following column (4) of Table 2, we set the wage elasticity to -0.719 and the tightness elasticity to -0.050 , which yields a ratio of 14.4. Using IEB data for our period of study, we further calculate a yearly separation rate δ of 26.8%.¹⁸ Building on evidence from business surveys (Jagannathan, Matsa, Meier, and Tarhan 2016; Graham 2022), we assume that firms’ yearly subjective discount rate r equals 15%. In line with empirical evidence, we set the elasticity of matches with respect to job seekers, ν , to 0.6 (Petrongolo and Pissarides 2001).

Table 4 presents alternative calibrations of Equation (11), which differ in how we model the elasticities of vacancy posting costs with respect to wages, ϕ_1 , and tightness, ϕ_2 . In column (1), we follow the canonical search-and-matching model with constant vacancy posting costs per period and set both elasticities to zero, implying that pre-match hiring costs amount to 27.7% of annual wage payments. In column (2), the magnitude of pre-match hiring

¹⁸To minimize bias from temporal aggregation (Nordmeier 2014), we estimate a day-to-day separation rate of 0.08543%, which translates into a yearly separation rate of 26.8%: $1 - (1 - 0.0008543)^{365.25} = 0.268$.

costs increases to 31.4% when, along the lines of Beaudry et al. (2018), assuming an elasticity of vacancy posting costs with respect to wages of unity.

In column (3), we rely on the evidence from Muehlemann and Strupler Leiser (2018) who, by leveraging panel data, provide the most convincing estimates on the effect of wages ($\phi_1 = 1.852$) and labor market tightness ($\phi_2 = 0.468$) on pre-match hiring costs. In this calibration, pre-match hiring costs amount to 17.7% of annual wage payments. In column (4), we use our estimates from our cross-sectional regression of the direct component of pre-match hiring costs on wages ($\phi_1 = 0.524$) and tightness ($\phi_2 = 0.127$), as given by column (2) of panel (a) of Table 1, and arrive at a value of 24.1%. In column (5), when using our effects of wages ($\phi_1 = 0.624$) and tightness ($\phi_2 = 0.148$) on overall pre-match hiring costs, as implied by column (2) of panel (a) of Table 1, the magnitude reduces slightly to 23.6%. In column (6), we also quantify the magnitude of pre-match hiring costs using our extended model, which additionally accounts for the process of wage formation (see Equation (A.18) in Appendix A.2). As given by column (1) of Table 3, we calibrate the effect of tightness on wages, β_3 , to 0.004. After setting the observed average tightness to 0.636 (see Appendix Table C2) and the replacement rate to 0.67 (as specified in the German legal Social Code Book III) in this model, the magnitude of pre-match hiring costs amounts to 16.4% of annual wage payments.

By and large, our preferred calibrations, as given by columns (3)–(6) in Table 4, imply that pre-match hiring costs are substantial and amount to 16–24% in relation to annual wage payments.

Search Externalities and Reallocation Effects

In the presence of hiring frictions (i.e., $\beta_2 < 0$), an employer's labor demand decision affects the hiring decision in other firms through search externalities. In such a setting, the feedback effect on labor market tightness will partly offset any first-round response in labor demand due to reallocation effects. As a consequence, the aggregate own-wage elasticity of labor demand (i.e., including these so-called search or congestion externalities) will be less negative than the own-wage elasticity of labor demand of the single firm to the extent that higher (lower) aggregate employment amplifies (reduces) hiring frictions (Beaudry et al. 2018).

To gauge the magnitude of the feedback effect, $\omega = \beta_4 \cdot \beta_2$, we estimate the following empirical version of the Beveridge-curve relationship, as given by Equation (14), for the years 2012–2019:

$$(22) \quad \Delta \ln \theta_{zt} = \tilde{\beta}_0 + \beta_4 \cdot \Delta \ln L_{zt} + \Delta \xi_{zt}$$

where $\beta_4 = \frac{1}{1-\nu} \cdot (1 - \frac{\partial \ln U}{\partial \ln L}) > 0$. Specifically, we estimate the impact of changes in aggregate employment of full-time workers, L_{zt} , on labor market tightness in a commuting zone, given by $\theta_{zt} = \frac{V_{zt}}{U_{zt}}$. To rule out bias from

Table 4. Magnitude of Pre-Match Hiring Costs

<i>Model</i> <i>Assumptions on</i> <i>functional form of</i> <i>vacancy posting cost:</i>	(1) <i>Baseline</i> <i>Standard</i> <i>DMP</i> <i>model</i>	(2) <i>Baseline</i> <i>Beaudry</i> <i>et al.</i> <i>(2018)</i>	(3) <i>Baseline</i> <i>Muehleemann /</i> <i>Struppler Leiser</i> <i>(2018)</i>	(4) <i>Baseline</i> <i>IAB-JVS</i> <i>direct</i> <i>hiring</i> <i>costs</i>	(5) <i>Baseline</i> <i>IAB-JVS</i> <i>overall</i> <i>hiring</i> <i>costs</i>	(6) <i>Extended</i> <i>Standard</i> <i>DMP</i> <i>model</i>	<i>Sources</i>
Elasticity of labor demand w.r.t. wages (β_1)	-0.719	-0.719	-0.719	-0.719	-0.719	-0.719	Table 2, column (4)
Elasticity of labor demand w.r.t. tightness (β_2)	-0.050	-0.050	-0.050	-0.050	-0.050	-0.050	Table 2, column (4)
Yearly separation rate (δ)	0.268	0.268	0.268	0.268	0.268	0.268	Own calculations
Yearly discount rate (r)	0.150	0.150	0.150	0.150	0.150	0.150	[a], [b]
Elasticity of matches w.r.t. job seekers (v)	0.600	0.600	0.600	0.600	0.600	0.600	[c]
Elasticity of hiring costs w.r.t. wages (ϕ_1)	0.000	1.000 ^[d]	1.852 ^[e]	0.524 ^[f]	0.624 ^[g]	0.000	[d], [e], [f], [g]
Elasticity of hiring costs w.r.t. tightness (ϕ_2)	0.000	0.000 ^[d]	0.468 ^[h]	0.127 ^[i]	0.148 ^[g]	0.000	[d], [f], [g], [h]
Elasticity of wages w.r.t. tightness (β_3)						0.004	Table 3, column (1)
Labor market tightness (θ)						0.636	Appendix Table C2
Replacement rate ($\overline{rr}$)						0.670	Social Code Book III
Pre-match hiring costs as fraction of annual wage payments ($\frac{c}{w}$)	0.277	0.314	0.177	0.241	0.236	0.164	Eqs. (11), (A.35)

Source: Own illustration.

Notes: The table displays the calibration of Equation (11) to approximate the magnitude of pre-match hiring costs. DMP = Diamond-Mortensen-Pissarides; IAB-JVS = IAB Job Vacancy Survey. [a] = Jagannathan et al. (2016, table 3); [b] = Graham (2022, table 2); [c] = Petrongolo and Pissarides (2001, p. 424); [d] = Beaudry et al. (2018, p. 2724); [e] = Muehleemann and Struppler Leiser (2018, table 8); [f] = Table 1, panel (a), column (2); [g] = Table 1, panel (a), column (6); [h] = 0.323 · 1.45 = Muehleemann and Struppler Leiser (2018, table 8, p. 125).

Table 5. Feedback Regression of Tightness on Aggregate (Un-)Employment

	(1) $\Delta \text{ Log regional tightness}$	(2) $\Delta \text{ Log regional job seekers}$
$\Delta \text{ Log regional employment}$	9.284*** (0.973)	-4.044*** (0.387)
Instruments	Z^L	Z^L
Observations	357	357
Clusters	51	51
$F: \Delta \text{ Log employment}$	154	154

Sources: Integrated Employment Biographies, Official Statistics from the Federal Employment Agency, IAB Job Vacancy Survey, 1999–2019.

Notes: The table displays IV regressions of differences in log labor market tightness per commuting zone on differences in log aggregate employment (of regular full-time workers) in the respective commuting zone. The instrumental variables (IV) refer to shift-share instruments of yearly national changes in occupations weighted by occupational employment in the respective commuting zone as of 1999. The lag difference is one year. Labor markets are combinations of 5-digit KldB occupations and commuting zones. Standard errors (in parentheses) are clustered at the commuting-zone level. $F = F$ -statistics of excluded instruments; KldB = German Classification of Occupations; L = employment; U = job seekers; V = vacancies; Z = shift-share instrument.

* = $p < 0.10$; ** = $p < 0.05$; *** = $p < 0.01$.

reverse causality, we construct a conventional Bartik instrument for employment at the regional level: $Z_{zt}^L = \sum_{o=1}^O \frac{L_{zo\tau}}{L_{z\tau}} \cdot \Delta \ln L_{ot}$. We set the base period τ at year 1999 to ensure predetermined employment shares and estimate Equation (22) in one-year differences.

Column (1) in Table 5 displays the IV results of the feedback effect. In line with theory, the feedback effect of aggregate employment on tightness, β_4 , turns out to be significantly positive. We find that a 1% increase in aggregate employment raises labor market tightness by 9.3%. Such a positive impact of aggregate employment on labor market tightness gives rise to a self-dampening feedback cycle when aggregate employment shifts.

To check the plausibility of this value, we estimate the impact of regional employment on the number of job seekers per commuting zone in an analogous specification to Equation (22). Column (2) in Table 5 shows that an increase in employment by 1% is associated with a reduction in job seekers by 4.0%. We solve β_4 for the elasticity of matches with respect to the number of job seekers, ν , and, when inserting the estimates from Table 5, arrive at a value of 0.54.¹⁹ Favorably, the implied matching elasticity is well in line with the empirical literature on the matching function in Germany (Fahr and Sunde 2006a, 2006b).

Finally, we insert our wage effect, β_1 , our tightness effect, β_2 , and our feedback effect, β_4 , into Equation (15) to calculate the aggregate own-wage elasticity of labor demand. Thus, by virtue of the self-correcting feedback

¹⁹ $\nu = 1 - \frac{1 - \frac{\Delta \ln U}{U}}{\beta_4} = 1 - \frac{1 - (-4.044)}{9.284} = 0.54$.

mechanism, the individual-firm own-wage elasticity of labor demand for full-time workers shrinks from -0.71 to -0.49 when accounting for search externalities at the aggregate level.²⁰ More generally speaking, by factoring in the feedback cycle, aggregate changes in labor demand (of whatever reason) shrink by 31.7% since the associated changes in labor market tightness facilitate reallocation effects between firms.²¹

The described mechanism can be helpful to reconcile evidence on the own-wage elasticity of labor demand with ex post evaluations of minimum wages: Despite evidence that firms reduce their labor demand when facing higher wages, the reallocation of workers to other firms can rationalize why minimum wages are frequently found to have only limited disemployment effects (Bossler and Gerner 2020; Dustmann et al. 2022).

Conclusion

We integrate components of the canonical search-and-matching model into a labor demand equation. Through the lens of our model, we estimate the effect of not only wages but also labor market tightness on firms' labor demand. To address issues of endogeneity, we construct novel shift-share instruments at the firm level. For the purpose of estimating our model, we leverage the universe of administrative employment records in Germany along with official statistics and survey data on vacancies and job seekers.

We report an own-wage elasticity of labor demand of -0.7 at the firm level. Further, our estimates imply that a doubling in labor market tightness, as observed between 2012 and 2019, reduces firms' employment by 5% in the medium run, highlighting that the tightening of labor markets amplifies hiring frictions. When accounting for the associated search externalities, the own-wage elasticity of labor demand reduces to -0.5 at the regional level. Thus, aggregate changes in labor demand are dampened by approximately 30% due to reallocation effects between firms. In light of our model, pre-match hiring costs amount to 16–24% of annual wage payments.

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²⁰ $\beta_1^{agg} = \frac{\beta_1}{1 - (\beta_1 \beta_2)} = \frac{-0.719}{1 - (-0.050 \cdot 9.284)} = -0.491$.

²¹ $1 - \frac{1}{1 - \omega} = 1 - \frac{1}{1 - (\beta_1 \beta_2)} = \frac{1}{1 - (-0.050 \cdot 9.284)} = 0.317$.

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